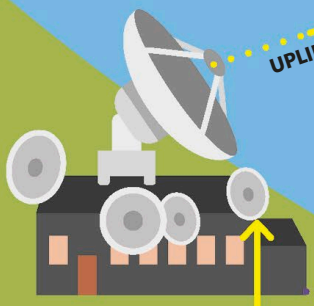


6 Communications satellite
A satellite receives radio signals transmitted from a ground-based station and relays them back to Earth to another exchange. Satellites are used infrequently for telephone calls due to the delay on the signals.

UPLINK

DOWNLINK



5 International exchange
An international telephone exchange links national telephone networks to the rest of the PSTN, allowing for communication across borders using international dialing codes.

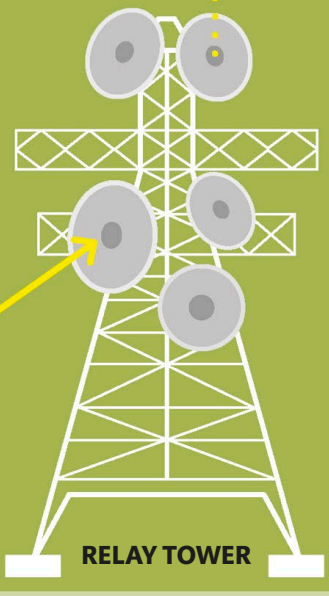


INTERNATIONAL EXCHANGE

UNDERWATER CABLE

INTERNATIONAL EXCHANGE

Undersea cables carry telephone traffic between land-based stations, which could be separated by entire oceans



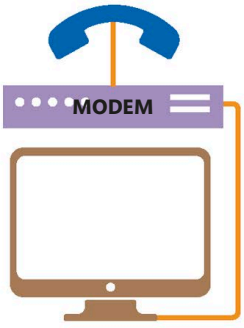
RELAY TOWER

4 Relay tower
Tall relay towers receive and retransmit signals in order to establish wireless telecommunications channels between distant telephone exchanges.

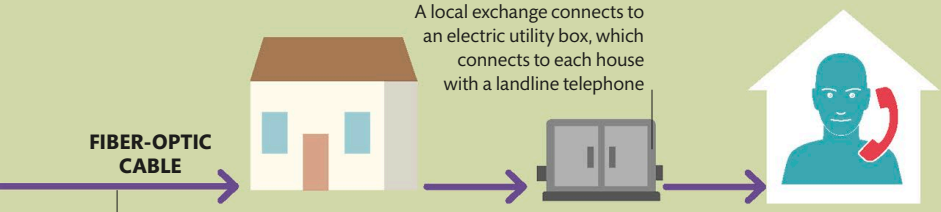
Telephone infrastructure
An international cell phone call and a long-distance landline call share most of the same infrastructure, including a main exchange. However, to cover the significant distances required, making an international call may involve transmitting signals by underwater cable or, rarely, radio waves, while many landline calls are made using just electric and fiber-optic cables.

DIAL-UP INTERNET

Dial-up is a form of Internet access that uses the PSTN. The user's computer sends information down the telephone line to the Internet, via his or her Internet service provider (ISP). This process requires a modem (short for modulator-demodulator) to encode and decode audio signals from the telephone line. Millions of people living in remote areas still rely on dial-up Internet.



A local exchange connects to an electric utility box, which connects to each house with a landline telephone



FIBER-OPTIC CABLE

LOCAL EXCHANGE

UTILITY BOX

Often, underground fiber-optic cables (see pp.190–191) connect main exchange to local exchange

4 Incoming landline call
When the call reaches its destination, the recipient's phone rings. When it is picked up, a connection is established and the conversation can begin (see pp.184–185).